



COSMOS AI at Scale

Language Model

July 2026, Instructor: Meng Tang

Many slides from Julia Hockenmaier and Christopher Manning

ChatGPT (Generative Pretrained Transformer)



You

What are top 10 things to do in Merced?



ChatGPT

Merced, California, offers a range of activities and attractions for visitors to enjoy. Here are the top 10 things to do in Merced:

1. **Merced National Wildlife Refuge:** Explore the refuge and observe diverse bird species, including migratory birds, waterfowl, and endangered species like the Aleutian Canada goose.
2. **Castle Air Museum:** Discover the history of aviation at this museum, which features a vast collection of aircraft, including military planes, helicopters, and memorabilia.
3. **Yosemite National Park:** While not directly in Merced, the park is nearby and offers breathtaking scenery, hiking trails, waterfalls, and opportunities for camping and wildlife viewing.
4. **Applegate Park and Zoo:** Enjoy a day outdoors at this park, which features playgrounds, picnic areas, a small zoo with animals like monkeys, birds, and reptiles, and a miniature train ride.
5. **Downtown Merced:** Explore the historic downtown area with its charming shops, cafes, restaurants, and galleries. Don't miss the Merced County Courthouse Museum to learn about the region's history.
6. **Merced County Fairgrounds:** Check out events and festivals held at the fairgrounds throughout the year, including the Merced County Fair, concerts, rodeos, and more.
7. **Lake Yosemite:** Spend a relaxing day by the lake, where you can enjoy fishing, boating, kayaking, picnicking, and walking along the scenic trails.

What is Natural Language Processing?

- ❑ Natural language processing is the set of methods for making human language accessible to computers.
(Jacob Eisenstein)
- ❑ Natural language processing is the field at the intersection of Computer science (Artificial intelligence) and linguistics.
(Christopher Manning)

NLP application: Machine translation

爷爷心疼孙女让娃睡懒觉 没想到引发了家庭矛盾

2020-08-25 08:06:03 来源: 钱江晚报

70岁的钟大伯(化名)陷入了“暑期焦虑”:这个暑假,他每周都要接送孙女上下培训班。高温、酷暑,每天回来,都像脚踩棉花般没力气。

除了身体上的不适,还有精神上的紧张。

觉得儿子儿媳给孩子报班太多,钟大伯还和他们产生了冲突:“大热天的,大人孩子都遭罪。”

这段时间,钟大伯因为容易激动上火,失眠,胃口差,血压一直不稳定,来到了浙江省人民医院精神卫生科就诊。

Google Translate

Grandpa feels sorry for his granddaughter and let the baby sleep in

2020-08-25 08:06:03 Source: Qianjiang Evening News

Uncle Zhong (a pseudonym), 70, fell into "summer anxiety": This summer, he would shuttle his granddaughter to and from training classes every week. With high temperatures and scorching heat, every day I come back, I feel as weak as stepping on cotton.

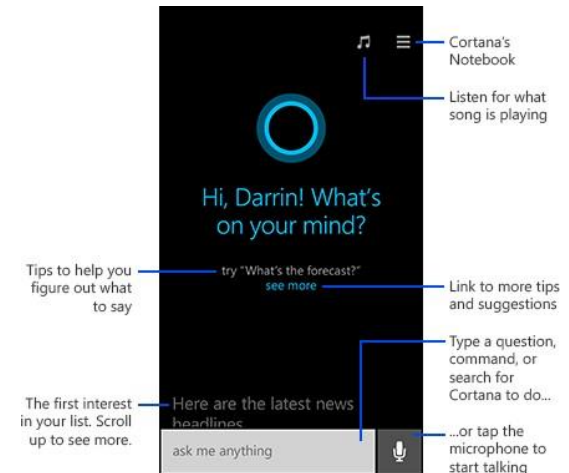
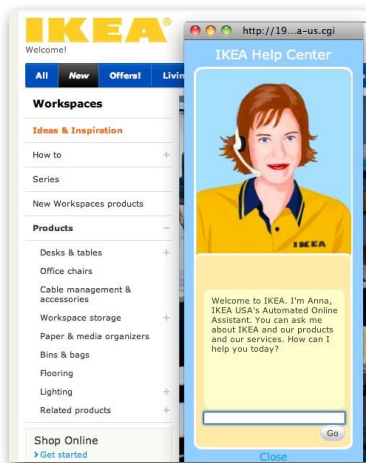
In addition to physical discomfort, there is also mental tension.

Feeling that his son and daughter-in-law were reporting too much for their children, Uncle Zhong also had a conflict with them: "It's a hot day, adults and children suffer."

During this period of time, Uncle Zhong came to the Mental Health Department of Zhejiang Provincial People's Hospital because he was prone to get angry, insomnia, poor appetite, and unstable blood pressure.

http://education.news.cn/2020-08/25/c_1210768533.htm

NLP application: Dialog systems, chatbots, assistants

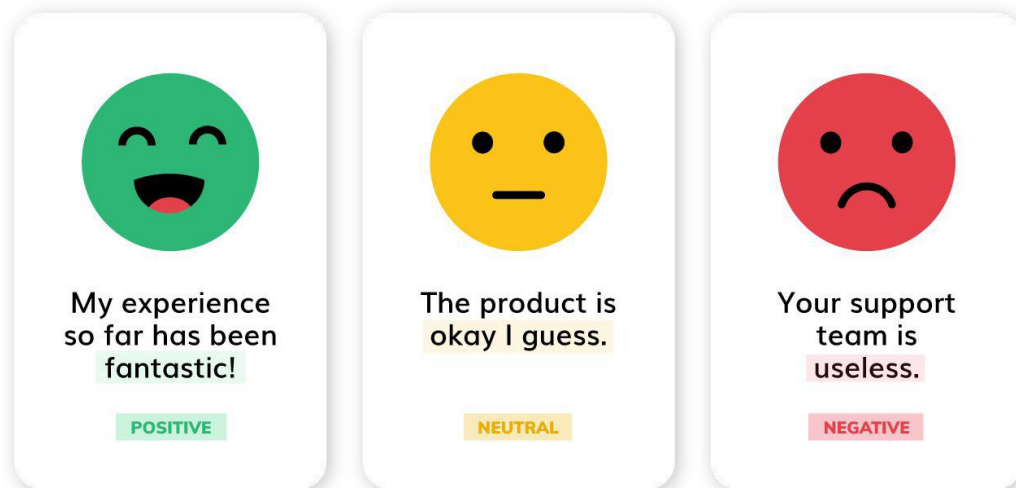


ChatGPT		
Examples	Capabilities	Limitations
"Explain quantum computing in simple terms" →	Remembers what user said earlier in the conversation	May occasionally generate incorrect information
"Got any creative ideas for a 10 year old's birthday?" →	Allows user to provide follow-up corrections	May occasionally produce harmful instructions or biased content
"How do I make an HTTP request in Javascript?" →	Trained to decline inappropriate requests	Limited knowledge of world and events after 2021



NLP application: Sentiment analysis

- Determine the meaning behind is positive, negative, or neutral



Outline

- ❑ Language Model
 - ❑ What is language model?
 - ❑ N-gram language model



Part I: Language Modeling

English Vocabulary

How large is the **vocabulary** of English (or any other language)?

Vocabulary size = the number of distinct word types

If you count words in text, you will find that...

...a **few words** are **very frequent**

(the, be, to, of, and, a, in, that,...)

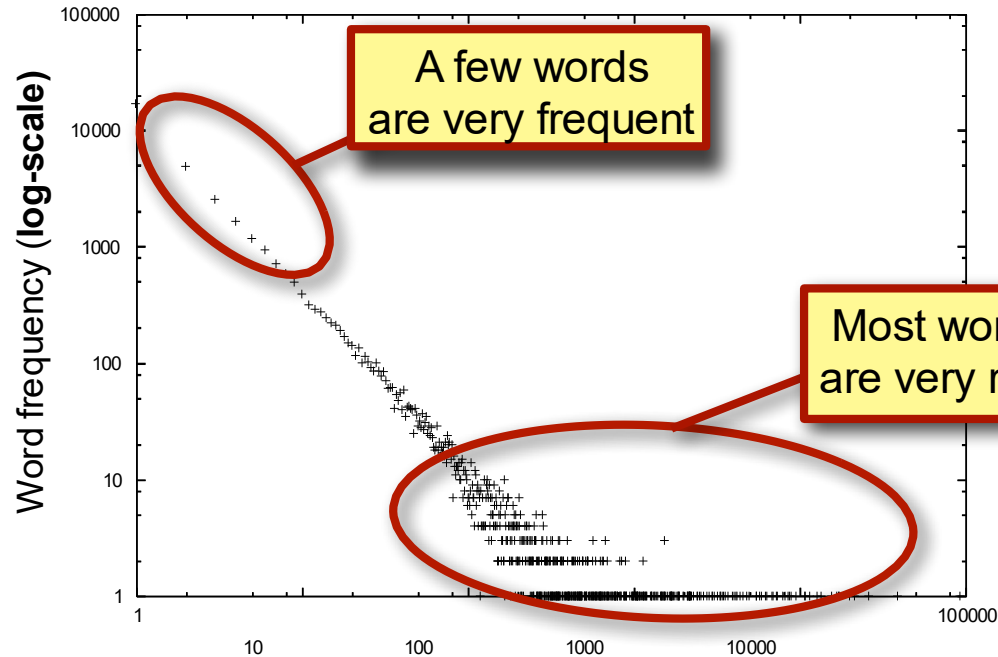
... **most words** are **very rare**.

... even if you've read a lot of text,

you will keep finding **words you haven't seen before**.

Word frequency: the number of occurrences of a word type in a text (or in a collection of texts)

Long-tailed word distribution



English words, sorted by frequency (**log-scale**)

$w_1 = \textit{the}$, $w_2 = \textit{to}$, ..., $w_{5346} = \textit{computer}$, ...

Why do we need language models?

Many NLP tasks require **natural language output**:

- Machine translation**: return text in the target language
- Speech recognition**: return a transcript of what was spoken
- Natural language generation**: return natural language text
- Spell-checking**: return corrected spelling of input

Language models define **probability distributions over (natural language) strings or sentences**.

- We can use a language model to **generate strings**
- We can use a language model to **score/rank candidate strings** so that we can choose the best (i.e. most likely) one:
if $P_{LM}(A) > P_{LM}(B)$, return A, not B

Now let's look at natural language

Text as a bag of words

Alice was beginning to get very tired of sitting by her sister on the bank, and of having nothing to do: once or twice she had peeped into the book her sister was reading, but it had no pictures or conversations in it, 'and what is the use of a book,' thought Alice 'without pictures or conversation?'

$$P(\text{of}) = 3/66$$

$$P(\text{Alice}) = 2/66$$

$$P(\text{was}) = 2/66$$

$$P(\text{to}) = 2/66$$

$$P(\text{her}) = 2/66$$

$$P(\text{sister}) = 2/66$$

$$P(,) = 4/66$$

$$P(') = 4/66$$

Sampling with replacement

A sampled sequence of words

beginning by, very Alice but was and?
reading no tired of to into sitting
sister the, bank, and thought of
without her nothing: having
conversations Alice once do or on she
it get the book her had peeped was
conversation it pictures or sister
in, 'what is the use had twice of a
book 'pictures or' to

$$P(\text{of}) = 3/66$$

$$P(\text{to}) = 2/66$$

$$P(,) = 4/66$$

$$P(\text{Alice}) = 2/66$$

$$P(\text{her}) = 2/66$$

$$P(') = 4/66$$

$$P(\text{was}) = 2/66$$

$$P(\text{sister}) = 2/66$$

In this model, $P(\text{English sentence}) = P(\text{word salad})$

Language modeling with N-grams

A **language model** over a vocabulary V assigns probabilities to strings drawn from V^* .

How do we compute the **probability of a string** $w^{(1)} \dots w^{(i)}$?

Recall the **chain rule**:

$$P(w^{(1)} \dots w^{(i)}) = P(w^{(1)}) \cdot P(w^{(2)} | w^{(1)}) \cdot \dots \cdot P(w^{(i)} | w^{(i-1)}, \dots, w^{(1)})$$

An **n-gram** language model assumes each word $w^{(i)}$ depends only on the **last n-1 words** $w^{(i-1)}, \dots, w^{(i-(n+1))}$

$$P_{ngram}(w^{(1)} \dots w^{(i)}) = P(w^{(1)}) \cdot P(w^{(2)} | w^{(1)}) \cdot \dots \cdot P(w^{(i)} | w^{(i-1)}, \dots, w^{(i-(n+1))})$$

N-gram models

N-gram models *assume* each word (event) depends only on the previous $n-1$ words (events):

$$\text{Unigram model: } P(w^{(1)} \dots w^{(N)}) = \prod_{i=1}^N P(w^{(i)})$$

$$\text{Bigram model: } P(w^{(1)} \dots w^{(N)}) = \prod_{i=1}^N P(w^{(i)} | w^{(i-1)})$$

$$\text{Trigram model: } P(w^{(1)} \dots w^{(N)}) = \prod_{i=1}^N P(w^{(i)} | w^{(i-1)}, w^{(i-2)})$$

Independence assumptions where the n -th event in a sequence depends only on the last $n-1$ events are called **Markov assumptions (of order $n-1$)**.

How many parameters do n-gram models have?

Given a vocabulary V of $|V|$ word types: so, for $|V| = 10^4$:

Unigram model: $|V|$ parameters 10^4 parameters

(one distribution $P(w^{(i)})$ with $|V|$ outcomes
[each $w \in V$ is one outcome])

Bigram model: $|V|^2$ parameters 10^8 parameters

Trigram model: $|V|^3$ parameters 10^{12} parameters

A bigram model for Alice

Alice was beginning to get very tired
of sitting by her sister on the bank,
and of having nothing to do: once or
twice she had peeped into the book
her sister was reading, but it had no
pictures or conversations in it, 'and
what is the use of a book,' thought
Alice 'without pictures or
conversation?'

$$P(w^{(i)} = \text{of} \mid w^{(i-1)} = \text{tired}) = 1$$

$$P(w^{(i)} = \text{of} \mid w^{(i-1)} = \text{use}) = 1$$

$$P(w^{(i)} = \text{sister} \mid w^{(i-1)} = \text{her}) = 1$$

$$P(w^{(i)} = \text{beginning} \mid w^{(i-1)} = \text{was}) = 1/2$$

$$P(w^{(i)} = \text{reading} \mid w^{(i-1)} = \text{was}) = 1/2$$

$$P(w^{(i)} = \text{bank} \mid w^{(i-1)} = \text{the}) = 1/3$$

$$P(w^{(i)} = \text{book} \mid w^{(i-1)} = \text{the}) = 1/3$$

$$P(w^{(i)} = \text{use} \mid w^{(i-1)} = \text{the}) = 1/3$$

Using a bigram model for Alice

English

Alice was beginning to get very tired of sitting by her sister on the bank, and of having nothing to do: once or twice she had peeped into the book her sister was reading, but it had no pictures or conversations in it, 'and what is the use of a book,' thought Alice 'without pictures or conversation?'

Word Salad

beginning by, very Alice but was and? reading no tired of to into sitting sister the, bank, and thought of without her nothing: having conversations Alice once do or on she it get the book her had peeped was conversation it pictures or sister in, 'what is the use had twice of a book 'pictures or' to

Now, $P(\text{English}) \gg P(\text{word salad})$

$$P(w^{(i)} = \text{of} \mid w^{(i-1)} = \text{tired}) = 1$$

$$P(w^{(i)} = \text{of} \mid w^{(i-1)} = \text{use}) = 1$$

$$P(w^{(i)} = \text{sister} \mid w^{(i-1)} = \text{her}) = 1$$

$$P(w^{(i)} = \text{beginning} \mid w^{(i-1)} = \text{was}) = 1/2$$

$$P(w^{(i)} = \text{reading} \mid w^{(i-1)} = \text{was}) = 1/2$$

$$P(w^{(i)} = \text{bank} \mid w^{(i-1)} = \text{the}) = 1/3$$

$$P(w^{(i)} = \text{book} \mid w^{(i-1)} = \text{the}) = 1/3$$

$$P(w^{(i)} = \text{use} \mid w^{(i-1)} = \text{the}) = 1/3$$

Learning (estimating) a language model

Where do we get the **parameters of our model** (its actual probabilities) from?

$$P(w^{(i)} = \text{'the'} \mid w^{(i-1)} = \text{'on'}) = ???$$

We need (a large amount of) text as **training data** to estimate the parameters of a language model.

The most basic parameter estimation technique:
relative frequency estimation (frequency = counts)

$$P(w^{(i)} = \text{'the'} \mid w^{(i-1)} = \text{'on'}) = C(\text{'on the'}) / C(\text{'on'})$$

Also called **Maximum Likelihood Estimation (MLE)**

$C(\text{'on the'})$ [or $f(\text{'on the'})$ for frequency]:

How often does 'on the' appear in the training data?

$$\text{NB: } C(\text{'on'}) = \sum_{w \in V} C(\text{'on' } w)$$

Handling unknown words: UNK

Training:

- Define a fixed vocabulary V such that all words in V appear at least n times in the training data
(e.g. all words that occur at least 5 times in the training corpus, or the most common 10,000 words in training)
- Add a new token UNK to V , and replace all other words in the corpus that are not in V by this token UNK
- Estimate the model on this modified training corpus.

Testing (when computing the probability of a string):

Replace any words not in the vocabulary by UNK

What about the beginning of the sentence?

In a trigram model

$$P(w^{(1)}w^{(2)}w^{(3)}) = P(w^{(1)})P(w^{(2)}|w^{(1)})P(w^{(3)}|w^{(2)},w^{(1)})$$

only the third term $P(w^{(3)}|w^{(2)},w^{(1)})$ is an actual trigram probability. What about $P(w^{(1)})$ and $P(w^{(2)}|w^{(1)})$?

Add **n-1 beginning-of-sentence (BOS)** symbols to each sentence for an **n-gram model**:

BOS₁ BOS₂ Alice was ...

Now the unigram and bigram probabilities involve only BOS symbols.

Summary: Estimating a bigram model with BOS (<s>), EOS (</s>) and UNK using MLE

1. Replace all words not in V in the training corpus with UNK
2. Bracket each sentence by special start and end symbols:
 <s> Alice was beginning to get very tired ...</s>
3. Define the Vocabulary V' = all tokens in modified training corpus
 (all common words, UNK, <s>, </s>)
4. Count the frequency of each bigram....
 $C(<s> \text{ Alice}) = 1, C(\text{Alice was}) = 1, \dots$
5. and normalize these frequencies to get probabilities:

$$P(\text{was} | \text{Alice}) = \sum_{w_i \in V'} \frac{C(\text{Alice was})}{C(\text{Alice } w_i)}$$

How do we use language models?

Independently of any application, we could use a language model as a *random sentence generator*

We can use a language model as a *sentence ranker*.

We prefer output sentences S_{Out} that have a higher language model probability. We can use a language model $P(S_{\text{Out}})$ to *score and rank these different candidate output sentences*, e.g. as follows:

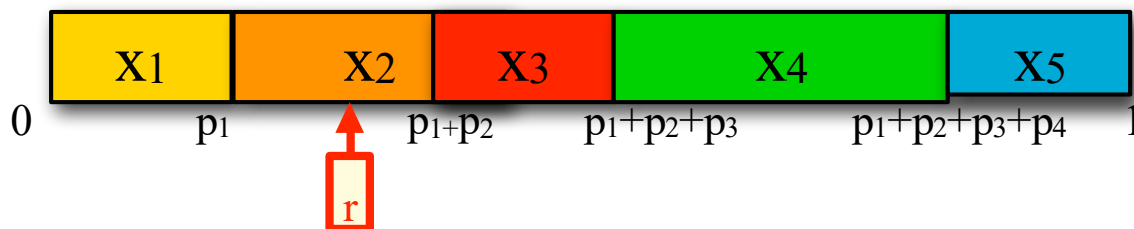
$$\operatorname{argmax}_{S_{\text{Out}}} P(S_{\text{Out}} \mid \text{Input}) = \operatorname{argmax}_{S_{\text{Out}}} P(\text{Input} \mid S_{\text{Out}})P(S_{\text{Out}})$$

Generating from a distribution

How do you generate text from an n -gram model?

That is, how do you sample from a distribution $P(X | Y=y)$?

- Assume X has N possible outcomes (values): $\{x_1, \dots, x_N\}$ and $P(X=x_i | Y=y) = p_i$
- Divide the interval $[0,1]$ into N smaller intervals according to the probabilities of the outcomes
- Generate a random number r between 0 and 1.
- Return the x_i whose interval the number is in.



Generating the Wall Street Journal

unigram: Months the my and issue of year foreign new exchange's september were recession exchange new endorsed a acquire to six executives

bigram: Last December through the way to preserve the Hudson corporation N. B. E. C. Taylor would seem to complete the major central planners one point five percent of U. S. E. has already old M. X. corporation of living on information such as more frequently fishing to keep her

trigram: They also point to ninety nine point six billion dollars from two hundred four oh six three percent of the rates of interest stores as Mexico and Brazil on market conditions

Generating Shakespeare

Unigram	• To him swallowed confess hear both. Which. Of save on trail for are ay device and rote life have • Every enter now severally so, let • Hill he late speaks; or! a more to leg less first you enter • Are where exeunt and sighs have rise excellency took of.. Sleep knave we. near; vile like
Bigram	• What means, sir. I confess she? then all sorts, he is trim, captain. • Why dost stand forth thy canopy, forsooth; he is this palpable hit the King Henry. Live king. Follow. • What we, hath got so she that I rest and sent to scold and nature bankrupt, nor the first gentleman? • Enter Menenius, if it so many good direction found'st thou art a strong upon command of fear not a liberal largess given away, Falstaff! Exeunt
Trigram	• Sweet prince, Falstaff shall die. Harry of Monmouth's grave. • This shall forbid it should be branded, if renown made it empty. • Indeed the duke; and had a very good friend. • Fly, and will rid me these news of price. Therefore the sadness of parting, as they say, 'tis done.
Quadrigram	• King Henry. What! I will go seek the traitor Gloucester. Exeunt some of the watch. A great banquet serv'd in; • Will you not tell me who I am? • It cannot be but so. • Indeed the short and the long. Marry, 'tis a noble Lepidus.

Shakespeare as corpus

The Shakespeare corpus has $N=884,647$ word tokens for a vocabulary of $V=29,066$ word types

Shakespeare used 300,000 bigram types out of $V^2= 844$ million possible bigram types.

99.96% of possible bigrams don't occur in this corpus.

MLE doesn't capture unseen events

We estimated a model on 884K word tokens, but:

Only 30,000 word types occur in the training data

Any word that does not occur in the training data has zero probability!

Only 0.04% of all possible bigrams (for 30K word types) occur in the training data

Our first neural net for NLP:

A neural n -gram model

Given a fixed-size vocabulary V , an n -gram model predicts the probability of the n -th word following the preceding $n-1$ words:

$$P(w^{(i)} | w^{(i-1)}, w^{(i-2)}, \dots, w^{i-(n-1)})$$

How can we model this with a neural net?

- **Input layer:** concatenate $n-1$ word vectors
- **Output layer:** a softmax over $|V|$ units

An n-gram model $P(w \mid w_1 \dots w_k)$ as a feedforward net (**naively**)

Assumptions:

The **vocabulary** V contains V types (incl. UNK, BOS, EOS)
We want to condition each word on k preceding words

Our (naive) model:

— [Naive]

Each **input word** $w_i \in V$ is a **V -dimensional one-hot vector** $v(w)$

→ The **input layer** $\mathbf{x} = [v(w_1), \dots, v(w_k)]$ has **$V \times k$ elements**

— We assume **one hidden layer** $\mathbf{h}$

— The **output layer** is a softmax over V elements

$$P(w \mid w_1 \dots w_k) = \text{softmax}(\mathbf{h}\mathbf{W}^2 + \mathbf{b}^2)$$

An n-gram model $P(w \mid w_1 \dots w_k)$ as a feedforward net (**better**)

Assumptions:

The **vocabulary** V contains V types (incl. UNK, BOS, EOS)
We want to condition each word on k preceding words

Our (better) model:

— [Better]

Each **input word** $w_i \in V$ is an **n -dimensional dense embedding vector** $v(w)$ (with $n \ll V$)

→ The **input layer** $\mathbf{x} = [v(w_1), \dots, v(w_k)]$ has $n \times k$ elements

— We assume **one hidden layer** $\mathbf{h}$

— The **output layer** is a softmax over V elements

$$P(w \mid w_1 \dots w_k) = \text{softmax}(\mathbf{h}\mathbf{W}^2 + \mathbf{b}^2)$$

Our neural n-gram models

Architecture:

Input Layer: $\mathbf{x} = [v(w_1) \dots v(w_k)]$

Hidden Layer: $\mathbf{h} = g(\mathbf{x}\mathbf{W}^1 + \mathbf{b}^1)$

Output Layer: $P(w \mid w_1 \dots w_k) = \text{softmax}(\mathbf{h}\mathbf{W}^2 + \mathbf{b}^2)$

How many parameters do we need? [# of weights and biases]:

Hidden layer with one-hot inputs: $\mathbf{W}^1 \in \mathbb{R}^{(k \cdot V) \times \dim(\mathbf{h})}$ $\mathbf{b}^1 \in \mathbb{R}^{\dim(\mathbf{h})}$

Hidden layer with dense inputs: $\mathbf{W}^1 \in \mathbb{R}^{(k \cdot n) \times \dim(\mathbf{h})}$ $\mathbf{b}^1 \in \mathbb{R}^{\dim(\mathbf{h})}$

Output layer (any inputs): $\mathbf{W}^2 \in \mathbb{R}^{\dim(\mathbf{h}) \times V}$ $\mathbf{b}^2 \in \mathbb{R}^V$

With $V = 10\text{K}$, $n = 300$ (word2vec), $\dim(\mathbf{h}) = 300$

$k = 2$ (trigram): $\mathbf{W}^1 \in \mathbb{R}^{20,000 \times 300}$ or $\mathbf{W}^1 \in \mathbb{R}^{600 \times 300}$ and $\mathbf{b}^1 \in \mathbb{R}^{300}$

$k = 5$ (six-gram): $\mathbf{W}^1 \in \mathbb{R}^{50,000 \times 300}$ or $\mathbf{W}^1 \in \mathbb{R}^{1500 \times 300}$ and $\mathbf{b}^1 \in \mathbb{R}^{300}$

$\mathbf{W}^2 \in \mathbb{R}^{300 \times 10,000}$ $\mathbf{b}^2 \in \mathbb{R}^{10,000}$

Six-gram model with one-hot inputs: 27,000,460,000 parameters,
with dense inputs: 3,460,000 parameters

Traditional six-gram model: $10^{4 \times 6} = 10^{24}$ parameters

1D CNNs for text

Text is a (variable-length) **sequence** of words (word vectors)

[#channels = dimensionality of word vectors]

We can use a **1D CNN** to slide a window of n tokens across:

— Filter size $n = 3$, stride = 1, no padding

The quick brown fox jumps over the lazy dog

The **quick brown fox** jumps over the lazy dog

The quick **brown fox jumps** over the lazy dog

The quick brown **fox jumps over** the lazy dog

The quick brown fox **jumps over the** lazy dog

The quick brown fox jumps **over the lazy** dog

— Filter size $n = 2$, stride = 2, no padding:

The quick brown fox jumps over the lazy dog

The quick **brown fox** jumps over the lazy dog

The quick brown fox **jumps over** the lazy dog

The quick brown fox jumps over **the lazy** dog

What we have learned today

- ❑ Language Modeling
- ❑ N-gram is a simple language model
- ❑ Neural N-gram Language Model
- ❑ Feedforward network and CNN for NLP
- ❑ Next lecture: Recurrent Neural Network, Sequence-t-sequence model, and transformer network